

OPEN UNIVERSITY OF SRI LANKA

Faculty of Natural Science

INDIVUDUAL REPORT

MINI PROJECT

(Group no:03)

OUSL STUDENT MAP

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Student ID: 124682005

Course code: COU4303

Course name: Artificial Intelligence

Academic Year : 2024/2025

Introduction

For the convenience of students studying at the Open University, we have developed a system using Prolog.This system informs students whether they are from the Colombo area or remote locations about the routes available to reach the university, while also providing rough estimates regarding security, travel time, and costs for each route.Once students register at the university, we enter their location into this system using their address.

Problem Statement

Students traveling to the university for essential academic activities may find their usual daily routes or modes of transport disrupted due to various natural events or other incidents.Therefore, they may be unable to arrive on time and could face severe difficulties.

For instance, consider a student from Matara who needs to attend a practical exam at 10:00 AM. Let us assume he usually travels to the university by train, but on this particular day, the train is delayed by an hour due to a fault on the railway line. Consequently, he might not be able to reach the university on time. He now needs to find a faster and more convenient route; furthermore, he must also consider the cost involved.

Solution

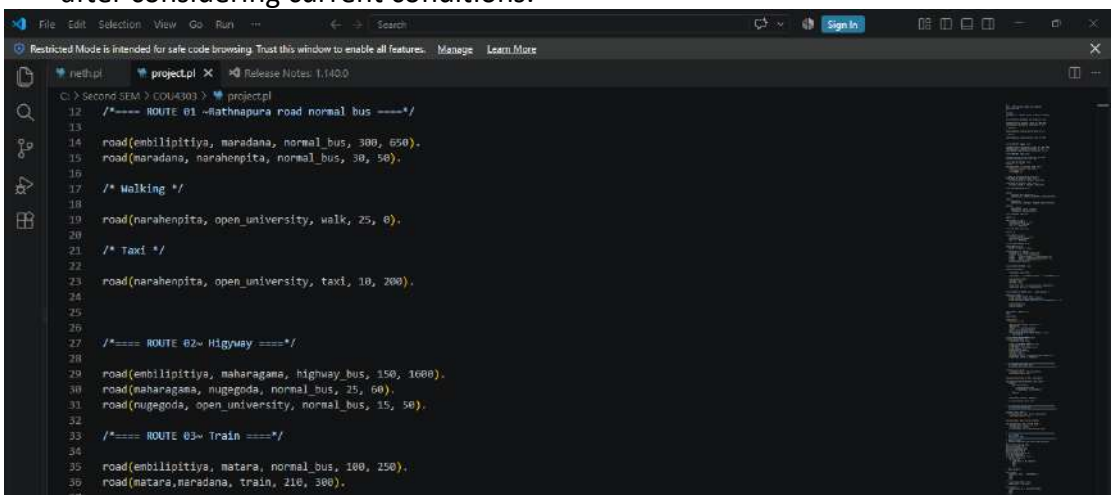
The system we have developed using Prolog provides solutions to the aforementioned issues. It incorporates three algorithms to help users identify routes that avoid impassable or blocked paths, enabling them to reach their destinations quickly and safely.We have used the A*, DFS, and BFS algorithms for this.

Upon logging into our system, students can view the routes and available modes of transport to travel from their current location to the university; most importantly, the system provides an approximate calculation of the travel time and cost.

Considering the example above, if the student has no financial constraints and prioritizes time, they could travel to Maharagama via an expressway bus and then take a taxi to the university. The system calculates and provides an estimate of the time and cost involved for this journey.

My Contribution

- Presenting a basic idea about the overall project and a rough outline for it.
- Coordinating the team and supervising the tasks assigned each members.
- Creating the basic structure for the code.
- Determining routes and inputting data into the code specially regarding transport mode costs and travel times for the path associated with each route after considering current conditions.



```
/*==== ROUTE 01 -Rathnapura road normal bus ====*/

road(embilipitiya, maradana, normal_bus, 300, 650).
road(maradana, narahenpita, normal_bus, 30, 50).

/* Walking */

road(narahenpita, open_university, walk, 25, 0).

/* Taxi */

road(narahenpita, open_university, taxi, 10, 200).



/*==== ROUTE 02- Higyway ====*/

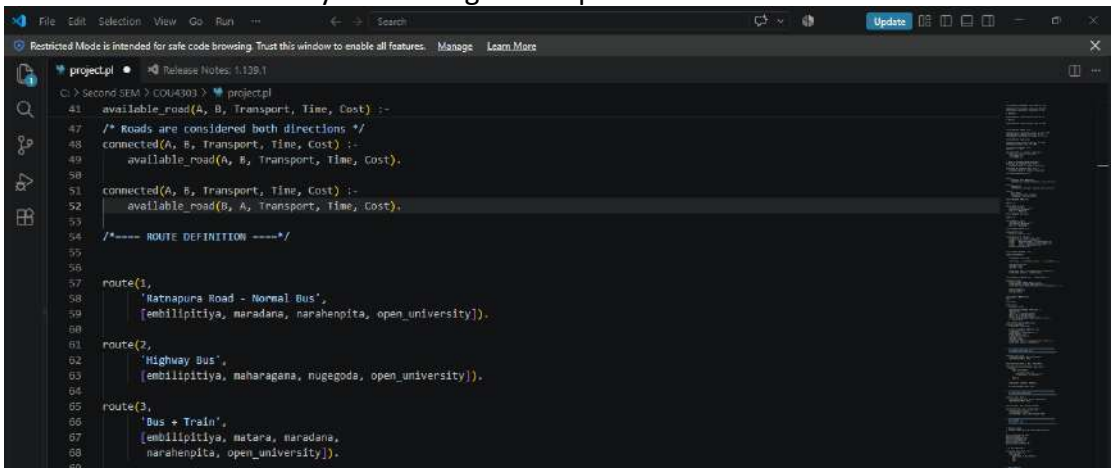
road(embilipitiya, maharagama, highway_bus, 150, 1600).
road(maharagama, nugegoda, normal_bus, 25, 60).
road(nugegoda, open_university, normal_bus, 15, 50).

/*==== ROUTE 03- Train ====*/

road(embilipitiya, matara, normal_bus, 100, 250).
road(matara,maradana, train, 210, 300).
```

- Adding comments to the code for ease of reading and understanding.

- Code was developed to interconnect routes and enable users to travel to their destinations using various modes of transport and routes from a single stop, while simultaneously calculating the respective travel times and costs.



```
available_road(A, B, Transport, Time, Cost) :-

/* Roads are considered both directions */
connected(A, B, Transport, Time, Cost) :-
    available_road(A, B, Transport, Time, Cost).

connected(A, B, Transport, Time, Cost) :-
    available_road(B, A, Transport, Time, Cost).

/*==== ROUTE DEFINITION ====*/


route(1,
      'Ratnapura Road - Normal Bus',
      [embilipitiya, maradana, narahenpita, open_university]).

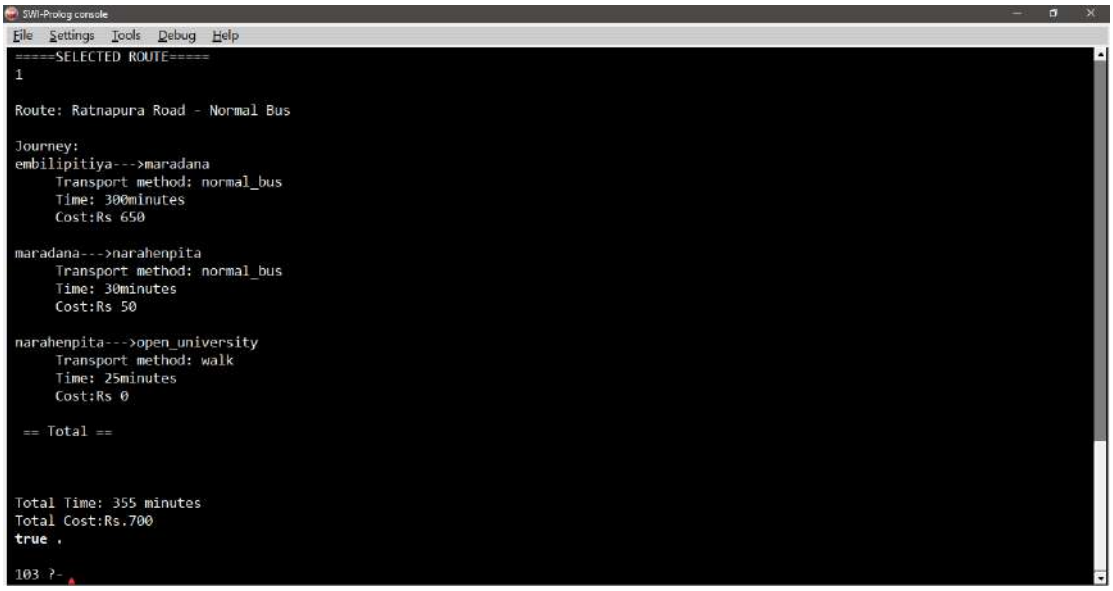
route(2,
      'Highway Bus',
      [embilipitiya, maharagama, nugegoda, open_university]).

route(3,
      'Bus + Train',
      [embilipitiya, matara, maradana,
       narahenpita, open_university]).
```

Conclusion

The code has been designed to find a path using the BFS and DFS algorithm and to handle repetitive execution.Writing code to find an efficient path using the A* algorithm.

They have been given the option to select and travel along one of the routes provided by the system that meets their requirements.



This system enables students to reach the university without unnecessary hassle by allowing them to consider the route, travel time, and cost based on their chosen mode of transport (public or private).